

Unit No 4	BACKUP, ARCHIVE AND REPLICATION	Lecture No	1
Topic	Introduction to Business Continuity		
Learning Outcome (LO) At the end of this lecture, students will be able to		Bloom's Knowledge Level	
LO1	Define Business Continuity	K1	
LO2	Explain Information Availability and causes of Information Unavailability.	K2	
LO3	Discuss the measures of Information Availability.	K2	

Introduction to Business Continuity

Business continuity (BC) is an integrated and enterprise wide process that includes all activities (internal and external to IT) that a business must perform to mitigate the impact of planned and unplanned downtime. BC entails preparing for, responding to, and recovering from a system outage that adversely affects business operations.

It involves proactive measures, such as business impact analysis and risk assessments, data protection, and security, and reactive countermeasures, such as disaster recovery and restart, to be invoked in the event of a failure. The goal of a business continuity solution is to ensure the “information availability” required to conduct vital business operations.

Information Availability

Information availability (IA) refers to the ability of the infrastructure to function according to business expectations during its specified time of operation. Information availability ensures that people (employees, customers, suppliers, and partners) can access information whenever they need it. Information availability can be defined with the help of reliability, accessibility and timeliness.

- **Reliability:** This reflects a component’s ability to function without failure, under stated conditions, for a specified amount of time.
- **Accessibility:** This is the state within which the required information is accessible at the right place, to the right user. The period of time during which the system is in an accessible state is termed *system uptime*; when it is not accessible it is termed *system downtime*.
- **Timeliness:** Defines the exact moment or the time window (a particular time of the day, week, month, and/or year as specified) during which information must be accessible. For example, if online access to an application is required between 8:00 am and 10:00 pm each day, any disruptions to data availability outside of this time slot are not considered to affect timeliness.

Causes of Information Unavailability

Various planned and unplanned incidents result in data unavailability. *Planned outages* include installation/integration/maintenance of new hardware, software upgrades or patches, taking backups, application and data restores, facility operations (renovation and construction), and refresh/migration of the testing to the production environment. *Unplanned outages include* failure caused by database corruption, component failure, and human errors.

Another type of incident that may cause data unavailability is natural or man-made disasters such as flood, fire, earthquake, and contamination. As illustrated in Figure 11-1, the majority of outages are planned. Planned outages are expected and scheduled, but still cause data to be unavailable. Statistically, less than 1 percent is likely to be the result of an unforeseen disaster.

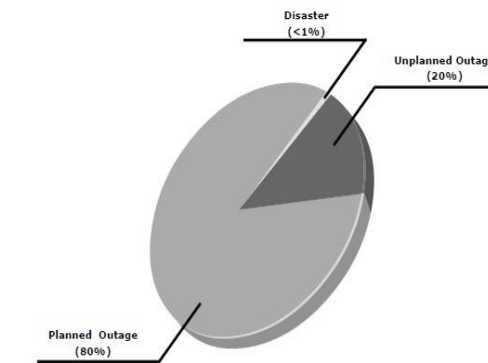


Figure 11-1: Disruptors of data availability

Measuring Information Availability

Information availability relies on the availability of the hardware and software components of a data center. Failure of these components might disrupt information availability. A failure is the termination of a component's ability to perform a required function.

The component's ability can be restored by performing an external corrective action, such as a manual reboot, a repair, or replacement of the failed component(s). Repair involves restoring a component to a condition that enables it to perform a required function within a specified time by using procedures and resources. Proactive risk analysis performed as part of the BC planning process considers the component failure rate and average repair time, which are measured by MTBF and MTTR:

- **Mean Time Between Failure (MTBF):** It is the average time available for a system or component to perform its normal operations between failures.
- **Mean Time To Repair (MTTR):** It is the average time required to repair a failed component. While calculating MTTR, it is assumed that the fault responsible for the failure is correctly identified and that the required spares and personnel are available. Note that a fault is a physical defect at the component level, which may result in data unavailability. MTTR includes the time required to do the following: detect the fault, mobilize the maintenance team, diagnose the fault, and obtain the spare parts, repair, test, and resume normal operations.

IA is the fraction of a time period that a system is in a condition to perform its intended function upon demand. It can be expressed in terms of system uptime and downtime and measured as the amount or percentage of system uptime:

$$IA = \text{system uptime} / (\text{system uptime} + \text{system downtime})$$

In terms of MTBF and MTTR, IA could also be expressed as

$$IA = MTBF / (MTBF + MTTR)$$

Consequences of Downtime

Data unavailability, or downtime, results in loss of productivity, loss of revenue, poor financial performance, and damages to reputation. Loss of productivity reduces the output per unit of labor, equipment, and capital. Loss of revenue includes direct loss, compensatory payments, future revenue losses, billing losses, and investment losses. Poor financial performance affects revenue recognition, cash flow, discounts, payment guarantees, credit rating, and stock price.

Damages to reputation may result in a loss of confidence or credibility with customers, suppliers, financial markets, banks, and business partners. Other possible consequences of downtime include the cost of additional equipment rental, overtime, and extra shipping. The business impact of downtime is the sum of all losses sustained as a result of a given disruption. An important metric, *average cost of downtime per hour*, provides a key estimate in determining the appropriate BC solutions.

It is calculated as follows:

Average cost of downtime per hour = average productivity loss per hour + average revenue loss per hour

Where:

Productivity loss per hour = (total salaries and benefits of all employees per week) / (average number of working hours per week)

Average revenue loss per hour = (total revenue of an organization per week) / (average number of hours per week that an organization is open for business)

The average downtime cost per hour may also include estimates of projected revenue loss due to other consequences such as damaged reputations and the additional cost of repairing the system.

Assessment questions to the lecture

Qn No	Question	Answer	Bloom's Knowledge Level
1.	Which of the following is NOT a common threat addressed by business continuity planning? a) Natural disasters b) Cyber-attacks c) Employee training programs d) Supply chain disruptions	c.	K1
2.	What does the term "information availability" refer to? a) The amount of data stored in a system b) The ease of access to information when needed c) The security measures implemented to protect information d) The process of sharing information with others	b.	K1

Students have to prepare answers for the following questions at the end of the lecture

Qn No	Question	Marks	CO	Bloom's Knowledge Level
1	What is Business Continuity?	2	4	K1
2	Discuss about Information Availability and the causes of Information Unavailability.	8	4	K2
3	Explain the measures of Information Availability.	8	4	K2

Reference Book

Author(s)	Title of the book	Page numbers
G Somasundaram Alok Shrivastava	EMC Corporation, Information Storage and Management, Wiley, India	229-233

Unit No 4	BACKUP, ARCHIVE AND REPLICATION	Lecture No	1
Topic	Backup and Recovery.		
Learning Outcome (LO) At the end of this lecture, students will be able to		Bloom's Knowledge Level	
LO1	Define Backup.	K1	
LO2	Define Recovery.	K1	
LO3	Differentiate Backup and Recovery.	K2	
LO4	Discuss the importance of Backup and Recovery in detail.	K2	

Backup and Recovery

A *backup* is a copy of production data, created and retained for the sole purpose of recovering deleted or corrupted data. With growing business and regulatory demands for data storage, retention, and availability, organizations are faced with the task of backing up an ever-increasing amount of data.

This task becomes more challenging as demand for consistent backup and quick restore of data increases throughout the enterprise which may be spread over multiple sites. Moreover, organizations need to accomplish backup at a lower cost with minimum resources.

Backups are performed to serve three purposes: disaster recovery, operational backup, and archival.

Disaster Recovery

Backups can be performed to address disaster recovery needs. The backup copies are used for restoring data at an alternate site when the primary site is incapacitated due to a disaster. Based on RPO and RTO requirements, organizations use different backup strategies for disaster recovery.

When a tape-based backup method is used as a disaster recovery strategy, the backup tape media is shipped and stored at an offsite location. These tapes can be recalled for restoration at the disaster recovery site. Organizations with stringent RPO and RTO requirements use remote replication technology to replicate data to a disaster recovery site. This allows organizations to bring up production systems online in a relatively short period of time in the event of a disaster.

Operational Backup

Data in the production environment changes with every business transaction and operation. *Operational backup* is a backup of data at a point in time and is used to restore data in the event of data loss or logical corruptions that may occur during routine processing.

The majority of restore requests in most organizations fall in this category. For example, it is common for a user to accidentally delete an important email or for a file to become corrupted, which can be restored from operational backup

Operational backups are created for the active production information by using incremental or differential backup techniques. An example of an operational backup is a backup performed for a production database just before a bulk batch update. This ensures the availability of a clean copy of the production database if the batch update corrupts the production database.

Archival

Backups are also performed to address archival requirements. Although CAS has emerged as the primary solution for archives, traditional backups are still used by small and medium enterprises for long-term preservation of transaction records, email messages, and other business records required for regulatory compliance.

Apart from addressing disaster recovery, archival, and operational requirements, backups serve as a protection against data loss due to physical damage of a storage device, software failures, or virus attacks. Backups can also be used to protect against accidents such as a deletion or intentional data destruction.

Backup Granularity

Most organizations use a combination of these three backup types to meet their backup and recovery requirements

- *Full backup* is a backup of the complete data on the production volumes at a certain point in time. A full backup copy is created by copying the data on the production volumes to a secondary storage device.
- *Incremental backup* copies the data that has changed since the last full or incremental backup, whichever has occurred more recently. This is much faster (because the volume of data backed up is restricted to changed data), but it takes longer to restore.
- *Cumulative (or differential) backup* copies the data that has changed since the last full backup. This method takes longer than incremental backup but is faster to restore.

Assessment questions to the lecture

Qn No	Question	Answer	Bloom's Knowledge Level
1.	Types of Backups ____, ____ and ____	Full Backup, Incremental Backup, Cumulative Backup	K1
2.	Backups are also performed to address ____ requirements. a. Physical b. Data c. Archival d. recovery	c	K1

3.	What is the primary goal of implementing backup and recovery procedures? a) To prevent unauthorized access to information b) To ensure the confidentiality of sensitive data c) To minimize data loss and maintain information availability d) To improve the performance of information systems	c.	K1
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Students have to prepare answers for the following questions at the end of the lecture

Qn No	Question	Marks	CO	Bloom's Knowledge Level
1	Define Backup	2	4	K1
2	Define Recovery	2	4	K1
3	Differentiate Backup and Recovery.	2	4	K2
4	Discuss the benefits of Backup and Recovery in details	16	4	K2

Reference Book

Author(s)	Title of the book	Page numbers
G Somasundaram Alok Shrivastava	EMC Corporation, Information Storage and Management, Wiley, India	251-256

Unit No 4	BACKUP, ARCHIVE AND REPLICATION	Lecture No	3
Topic	Backup targets and methods-Backup Process and Architecture- Backup and Restore Operations		
Learning Outcome (LO) At the end of this lecture, students will be able to		Bloom's Knowledge Level	
LO1	Discuss the backup methods and targets	K1	
LO2	Explain the Backup process and Architecture.	K2	
LO3	Explain Backup and Restore operations.	K2	
LO4	Differentiate Backup and Restore.	K2	

Backup Methods

Hot backup and cold backup are the two methods deployed for backup. They are based on the state of the application when the backup is performed. In a *hot backup*, the application is up and running, with users accessing their data during the backup process. In a *cold backup*, the application is not active during the backup process.

The backup of online *production data* becomes more challenging because data is actively being used and changed. An open file is locked by the operating system and is not copied during the backup process until the user closes it. The backup application can back up open files by retrying the operation on files that were opened earlier in the backup process.

During the backup process, it may be possible that files opened earlier will be closed and a retry will be successful. The maximum number of retries can be configured depending on the backup application. However, this method is not considered robust because in some environments certain files are always open.

Backup Process

A backup system uses client/server architecture with a backup server and multiple backup clients. The backup server manages the backup operations and maintains the backup catalog, which contains information about the backup process and backup metadata.

The backup server depends on backup clients to gather the data to be backed up. The backup clients can be local to the server or they can reside on another server, presumably to back up the data visible to that server. The backup server receives backup metadata from the backup clients to perform its activities.

The storage node is responsible for writing data to the backup device (in a backup environment, a storage node is a host that controls backup devices). Typically, the storage node is integrated with the backup server and both are hosted on the same physical platform. A backup device is attached directly to the storage node's host platform.

Some backup architecture refers to the storage node as the *media server* because it connects to the storage device. Storage nodes play an important role in backup planning because they can be used to consolidate backup servers.

The backup process is based on the policies defined on the backup server, such as the time of day or completion of an event. The backup server then initiates the process by sending a request to a backup client (backups can also be initiated by a client).

This request instructs the backup client to send its metadata to the backup server, and the data to be backed up to the appropriate storage node. On receiving this request, the backup client sends the metadata to the backup server. The backup server writes this metadata on its metadata catalog.

The backup client also sends the data to the storage node, and the storage node writes the data to the storage device. After all the data is backed up, the storage node closes the connection to the backup device. The backup server writes backup completion status to the metadata catalog.

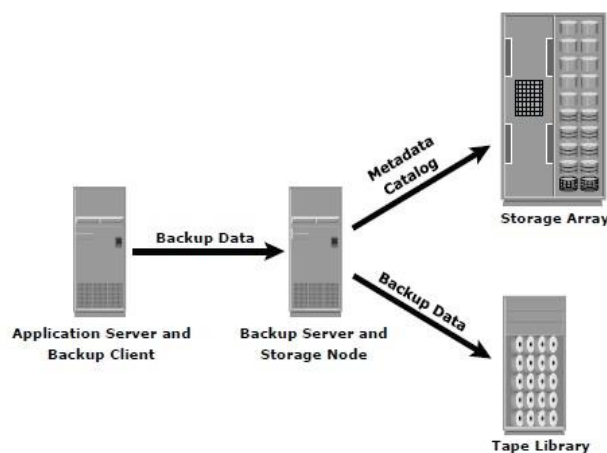


Figure 12-4: Backup architecture and process

Backup software also provides extensive reporting capabilities based on the backup catalog and the log files. These reports can include information such as the amount of data backed up, the number of completed backups, the number of incomplete backups, and the types of errors that may have occurred. Reports can be customized depending on the specific backup software used.

Backup and Restore Operations

When a backup process is initiated, significant network communication takes place between the different components of a backup infrastructure. The backup server initiates the backup process for different clients based on the backup schedule configured for them. For example, the backup process for a group of clients may be scheduled to start at 3:00 am every day.

The backup server coordinates the backup process with all the components in a backup configuration. The backup server maintains the information about backup clients to be contacted and storage nodes to be used in a backup operation. The backup server retrieves the backup-related information from the backup catalog and, based on this information, instructs the storage node to load the appropriate backup media into the backup devices.

Simultaneously, it instructs the backup clients to start scanning the data, package it, and send it over the network to the assigned storage node. The storage node, in turn, sends metadata to the backup server to keep it updated about the media being used in the backup process. The backup server continuously updates the backup catalog with this information.

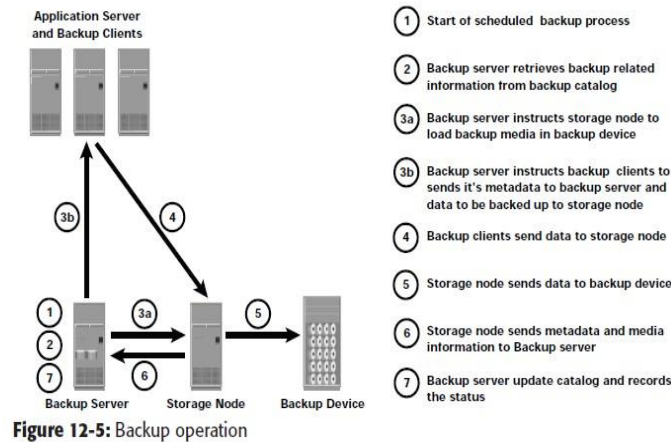


Figure 12-5: Backup operation

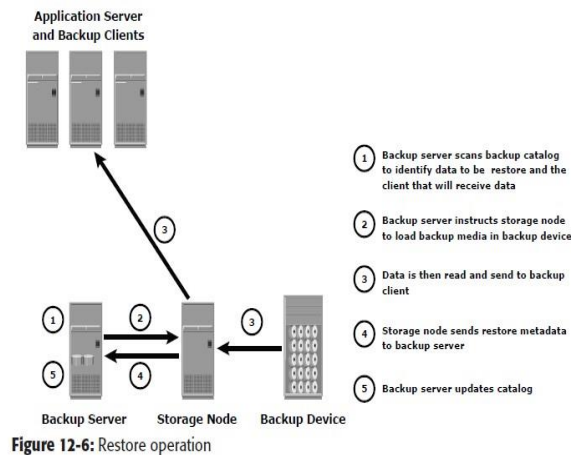


Figure 12-6: Restore operation

Upon receiving a restore request, an administrator opens the restore application to view the list of clients that have been backed up. While selecting the client for which a restore request has been made, the administrator also needs to identify the client that will receive the restored data. Data can be restored on the same client for whom the restore request has been made or on any other client.

The administrator then selects the data to be restored and the specified point in time to which the data has to be restored based on the RPO. Note that because all of this information comes from the backup catalog, the restore application must also communicate to the backup server.

The administrator first selects the data to be restored and initiates the restore process. The backup server, using the appropriate storage node, then identifies the backup media that needs to be mounted on the backup devices. Data is then read and sent to the client that has been identified to receive the restored data.

Some restorations are successfully accomplished by recovering only the requested production data. For example, the recovery process of a spreadsheet is completed when the specific file is restored. In database restorations, additional data such as log files and production data must be restored. This ensures application consistency for the restored data. In these cases, the RTO is extended due to the additional steps in the restoration process.

Assessment questions to the lecture

Qn No	Question	Answer	Bloom's Knowledge Level
1.	Which of the following is NOT a common backup target? a) External hard drive b) Cloud storage c) Main memory d) Tape drive	c.	K1
2.	Which component of the backup architecture is responsible for managing and executing backup operations? a) Backup client b) Backup server c) Backup storage d) Backup application	b.	K1
3	Which factor is crucial for determining the speed of a restore operation? a) Backup encryption b) Compression ratio c) Backup storage type d) Backup size	c.	K1

Students have to prepare answers for the following questions at the end of the lecture

Qn No	Question	Marks	CO	Bloom's Knowledge Level
1	List the backup methods.	2	2	K1
2	Differentiate Backup and Restore.	2	2	K1
3	Explain the Backup process and Architecture.	8	2	K2
4	Explain Backup and Restore operations.	8	2	K2
5	Illustrate the significance of backup and their dominant role in the business continuity of an organization. (April 2024)	16	2	K2

Reference Book

Author(s)	Title of the book	Page numbers
G Somasundaram Alok Shrivastava	EMC Corporation, Information Storage and Management, Wiley, India	258-262

Unit No 4	BACKUP, ARCHIVE AND REPLICATION	Lecture No	4
Topic	Data Deduplication		
Learning Outcome (LO) At the end of this lecture, students will be able to		Bloom's Knowledge Level	
LO1	Define Data Deduplication.	K1	
LO2	Describe the techniques to deduplicate data	K2	
LO3	Explain the process of data deduplication	K2	

Data Deduplication

Data deduplication is a process in which identical data is identified and only one instance of identical data is stored. Other instances of identical data are referenced to the stored instance. This reduces the amount of stored data.

The identification process might utilise cryptographic hashes such as MD5 or SHA to identify two identical sets of data. The deduplication process must maintain a list of identity characteristics (such as cryptographic hashes) for the data that has been stored. The process must also maintain a list of references for identical sets of data.

Data deduplication emerged as a key technology to dramatically reduce the amount of space and the cost that are associated with storing large amounts of data. Data deduplication is the art of intelligently reducing storage needs in order of magnitude. This method is better than common data compression techniques. Data deduplication works through the elimination of redundant data so that only one instance of a data set is stored.

IBM has the broadest portfolio of data deduplication solutions in the industry, which gives IBM the freedom to solve client issues with the most effective technology. Whether it is source or target, inline or post, hardware or software, disk or tape, IBM has a solution with the technology that best solves the problem:

- IBM ProtecTIER® Gateway and Appliance
- IBM System Storage N series Deduplication
- IBM Tivoli Storage Manager

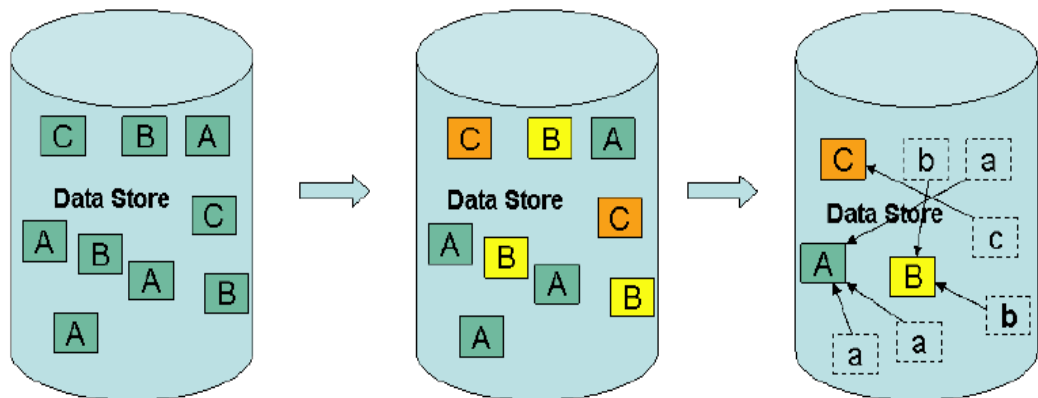
Data deduplication is a technology that reduces the amount of space that is required to store data on disk. It achieves this space reduction by storing a single copy of data that is backed up repetitively.

Data deduplication products read data while they look for duplicate data. Data deduplication products break up data into elements and create a signature or identifier for each data element. Then, they compare the data element signature to identify duplicate data. After they identify duplicate data, they retain one copy of each element. They create pointers for the duplicate items, and discard the duplicate items.

The effectiveness of data deduplication depends on many variables, including the rate of data change, the number of backups, and the data retention period. For example, if you back up the same incompressible data one time a week for six months, you save the first copy and you do not save the next 24. This method provides a 25:1 data deduplication ratio.

If you back up an incompressible file on week one, back up the exact same file again on week two, and never back it up again, this method provides a 2:1 data deduplication ratio.

A more likely scenario is that a portion of your data changes from backup to backup so that your data deduplication ratio changes over time. With data deduplication, you can minimize your storage requirements. Data deduplication can provide greater data reduction and storage space savings than other existing technologies.



1. Data elements (whole files, chunks, segments, streams, or DBs) are evaluated to determine a unique signature.

2. Signature values are compared to identify duplicates.

3. Duplicate data elements are replaced with pointers to a single stored element, saving storage space.

Data deduplication can reduce your storage requirements but the benefit you derive is determined by your data and your backup policies. Workloads with high database content have the highest data deduplication ratios.

However, product functions, such as IBM Tivoli Storage Manager Progressive Incremental or Oracle Recovery Manager (RMAN), can reduce the data deduplication ratio. Compressed, encrypted, or otherwise scrambled workloads typically do not benefit from data deduplication.

Good candidates for data deduplication are text files, log files, uncompressed and non-encrypted database files, email files (PST, DBX, and IBM Domino®), and Snapshots (Filer Snaps, BCVs, and VMware images).

Types of data deduplication and IBM HyperFactor Many vendors offer data deduplication products.

Various methods are available to deduplicate data. The following three methods are used frequently for data deduplication:

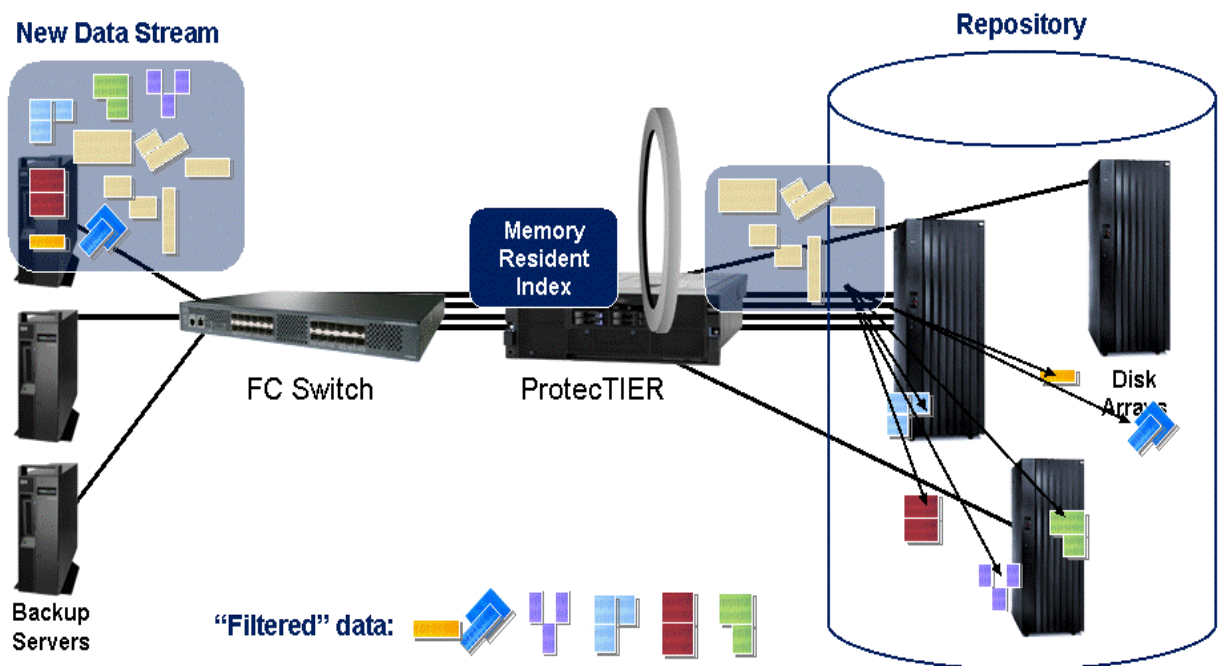
_ Hash-based data deduplication uses a hashing algorithm to identify chunks of data. Secure Hash Algorithm 1 (SHA-1) or Message-Digest Algorithm 5 (MDA-5) is commonly used. The details of each technique are beyond the intended scope of this publication.

_ Content-aware data deduplication methods are aware of the structure of common patterns of data that is used by applications. The content-aware data deduplication method assumes that the best candidate to deduplicate against is an object with the same properties, such as a file name. When a file match is identified, a bit-by-bit comparison is performed to determine whether data changed and the changed data is saved.

_ IBM HyperFactor® is a patented technology that is used in the IBM System Storage ProtecTIER Enterprise Edition and higher software. HyperFactor takes an approach that reduces the phenomenon of missed factoring opportunities, providing a more efficient process. With this approach, HyperFactor can surpass the reduction ratios that are attainable by any other data reduction method.

HyperFactor can reduce any duplicate data, regardless of its location or how recently it was stored. HyperFactor data deduplication uses a 4 GB Memory Resident Index to track similarities for up to 1 petabyte (PB) of physical disk in a single repository. HyperFactor technology uses a pattern algorithm that can reduce the amount of space that is required for storage by up to a factor of 25, based on evidence from existing implementations.

The capacity expansion that results from data deduplication is often expressed as a ratio, essentially the ratio of nominal data to the physical storage that is used.



Data deduplication processing

Data deduplication can either occur while the data is backed up to the storage media (real-time or *inline*) or after the data is written to the storage media (post-processing). Each method contains positive and negative aspects. These considerations must be evaluated by the engineer or technical specialist that is responsible for the concrete solution architecture and deployment.

IBM decided to use inline data deduplication processing because it offers larger target storage space without any need of a temporary disk cache pool for post-processed deduplication data. Bit comparison techniques, such as the technique that is used by ProtecTIER, were designed to provide 100% data integrity by avoiding the risk of hash collisions.

Assessment questions to the lecture

Qn No	Question	Answer	Bloom's Knowledge Level
1.	Duplication of Data is known as A. Data Repentance B. Data Redundancy C. Data Inconsistency D. Data base	b.	K1
2.	What is data deduplication in the context of data backup? A. The process of encrypting data B. The process of removing duplicate copies of data to save storage space C. The process of creating full backups D. The process of data recovery	b.	K1

Students have to prepare answers for the following questions at the end of the lecture

Qn No	Question	Marks	CO	Bloom's Knowledge Level
1	List the backup methods.	2	2	K1
2	Differentiate Backup and Restore.	2	2	K1
3	Explain the Backup process and Architecture.	8	2	K2
4	Explain Backup and Restore operations.	8	2	K2
5	Narrate the importance and types of data deduplication. (April 2024)	2	2	K1

Reference Book

Author(s)	Title of the book	Page numbers
Jon Tate Pall Beck Hector Hugo Ibarra Shanmuganathan Kumaravel Libor Miklas	Introduction to Storage Area Networks	146-149

Unit No 4	BACKUP, ARCHIVE AND REPLICATION	Lecture No	5
Topic	Cloud-based backup		
Learning Outcome (LO) At the end of this lecture, students will be able to		Bloom's Knowledge Level	
LO1	Define Cloud based backup.	K1	
LO2	Discuss Cloud backup use cases and approaches.	K2	

Cloud Based Backup:

Cloud backup, also known as *online backup* or *remote backup*, is a strategy for sending a copy of a physical or virtual file or database to a secondary, off-site location for preservation in case of equipment failure, site catastrophe or human malfeasance. The backup server and data storage systems are usually hosted by a third-party cloud or SaaS provider that charges the backup customer a recurring fee based on storage space or capacity used, data transmission bandwidth, number of users, number of servers or number of times data is retrieved.

Implementing cloud data backup can help bolster an organization's data protection, business continuance and regulatory compliance strategies without increasing the workload of IT staff. The labor-saving benefit can be significant and enough of a consideration to offset some of the additional costs associated with cloud backup, such as data transmission charges.

Most cloud subscriptions run on a monthly or yearly basis. Initially used mainly by consumers and home offices, online backup services are now commonly used by SMBs and larger enterprises to back up some forms of data. For larger companies, cloud data backup can serve as a supplementary form of backup.

What is the cloud?

Cloud computing is a general term that refers to hosted resources and services that are delivered over the internet. Different from traditional web hosting, the services on the cloud are sold on demand, offered in an elastic manner -- meaning the customer can use as much or as little of the service as needed -- and managed completely by the service provider. Additionally, a cloud can be private or public. A public cloud sells services to anyone on the internet, such as how AWS operates, while a private cloud supplies hosted services to a limited number of users within the business.

Cloud backup use cases and approaches

In an organization's data center, a backup application copies data and stores it on different media or another storage system for easy access in the event of a recovery situation. Although there are multiple options and approaches to off-site backup, cloud backup serves as the off-site facility for many organizations.

In an enterprise, the company might own the off-site server if it hosts its own private cloud service, but the chargeback method would be similar if the company uses a service provider to manage the cloud backup environment and receives a regular bill for backup storage and services. There are a variety of approaches to cloud backup, with available services that can easily fit into an organization's existing data protection process. Varieties of cloud backup include the following:

- **Backing up directly to the public cloud.** One way to store organizational workloads is by duplicating resources in the public cloud. This method entails writing data directly to cloud providers, such as AWS, Google Cloud or Microsoft Azure. The

organization uses its own backup software to create the data copy to send to the cloud storage service. The cloud storage service then provides the destination and safekeeping for the data, but it doesn't specifically provide a backup application. In this scenario, it's important that the backup software is capable of interfacing with the cloud's storage service. Additionally, with public cloud options, IT professionals might need to look into supplemental data protection procedures, such as data encryption as well as identity and access management to secure backed up data.

- **Backing up to a service provider.** In this scenario, an organization writes data to a cloud service or SaaS provider that offers backup services in a managed data center. The backup software that the company uses to send its data to the service might be provided as part of the service, or the service might support specific commercially available backup applications.
- **Choosing a cloud-to-cloud (C2C) backup.** These services are among the newest offerings in the cloud backup arena. They specialize in backing up data that already lives in the cloud, either as data created using a SaaS application or as data stored in a cloud backup service. As its name suggests a C2C backup service copies data from one cloud to another cloud. The cloud-to-cloud backup service typically hosts the software that handles this process.
- **Using online cloud backup systems.** There are also hardware alternatives that facilitate backing up data to a cloud backup service. These appliances are all-in-one backup machines that include backup software and disk capacity, along with the backup server. The appliances are about as close to plug-and-play as backup gets, and most of them also provide a seamless link to one or more cloud backup services or cloud providers. The list of vendors that offer backup appliances that include cloud interfaces is long, with Quantum, Unitrends, Arcserve, Rubrik, Cohesity, Dell EMC, StorageCraft and Asigra active in this arena. These appliances typically retain the most recent backup locally, in addition to copying it to the cloud backup provider, so any required recoveries can be made from the local backup copy, saving time and transmission costs.

Although the steps can vary based on backup method or type, this is the basic process for cloud backups. When an organization engages a cloud backup service, the first step is to complete a full backup of the data that must be protected. This initial backup can sometimes take days to finish uploading over a network as a result of the large volume of data being transferred.

In a 3-2-1 backup strategy, where an organization has three copies of data on two different media, at least one copy of the backed up data should be sent to an off-site backup facility so that it's accessible even if on-site systems are unavailable. Using a technique called *cloud seeding*, a cloud backup vendor sends a storage device -- such as a hard drive or tape cartridge -- to its new customer, which then backs up the data locally onto the device and returns it to the provider.

This process removes the need to send the initial data over the network to the backup provider. One example of a device that employs this technique is AWS Snowball Edge. If the amount of data in the initial backup is substantial, the cloud backup service might provide a full storage array for the seeding process.

These arrays are typically small network-attached storage (NAS) devices that can be shipped back and forth relatively easily. After the initial seeding, only changed data is backed up over the network.

How data is restored

Cloud backup services are typically built around a client software application that runs on a schedule determined by the purchased level of service and the customer's requirements. For example, if the customer has contracted for daily backups, the application collects, compresses, encrypts and transfers data to the cloud service provider's servers every 24 hours.

To reduce the amount of bandwidth consumed and the time it takes to transfer files, the service provider might only provide incremental backups after the initial full backup. Cloud backup services often include the software and hardware necessary to protect an organization's data, including applications for Microsoft Exchange and SQL Server.

Whether a customer uses its own backup application or the software the cloud backup service provides, the organization uses that same application to restore backed up data. Restorations could be on a file-by-file basis, by volume or a full restoration of the complete backup. More granular file-by-file restoration is typically the preferred method because it enables a business to quickly recover individual lost or damaged files rather than take the time and risk in restoring entire volumes.

If the volume of data to be restored is very large, the cloud backup service might ship the data on a complete storage array that the customer can hook up to its servers to recover its data. This is, in effect, a reverse seeding process. Restoring a large amount of data over a network can take an unacceptably long time, depending on the organization's recovery time objective.

A key feature of cloud backup restorations is that they can be done anywhere, from nearly any kind of computer. For example, an organization could recover its data directly to a disaster recovery site in a different location if its primary data center is unavailable.

Assessment questions to the lecture

Qn No	Question	Answer	Bloom's Knowledge Level
1.	What is cloud-based backup? a) Storing data on physical backup tapes b) Backing up data to a remote server over the internet c) Copying data to an external hard drive d) Replicating data on-premises	b.	K1
2.	Cloud computing is a general term that refers to _____ that are delivered over the internet.	hosted resources and services	K1
3.	Cloud backup services often include _____ necessary to protect an organization's data,	the software and hardware	K1

Students have to prepare answers for the following questions at the end of the lecture

Qn No	Question	Marks	CO	Bloom's Knowledge Level
1	Define Cloud based backup.	2	2	K1
2	Discuss Cloud backup usecases and approaches.	8	2	K1
3	Write the key aspects of Business continuity in the cloud (April 2024)	2	2	K1

Reference Book

Author(s)	Title of the book	Page numbers
	Web resources	

Unit No 4	BACKUP, ARCHIVE AND REPLICATION	Lecture No	6
Topic	Mobile device backup		
Learning Outcome (LO) At the end of this lecture, students will be able to		Bloom's Knowledge Level	
LO1	What are the key aspects of mobile device backup	K1	
LO2	Explain mobile device backup in detail.	K2	

Mobile Device Backup:

Mobile device backup involves creating copies of data stored on smart phones, tablets, or other mobile devices to prevent data loss in case of device damage, loss, or theft.

Key aspects of mobile device backup include:

- **Data Types:** Mobile device backup typically includes contacts, photos, videos, messages, app data, and other user-generated content.
- **Backup Methods:** Mobile device backup can be performed using various methods, including built-in backup features provided by the device's operating system, third-party backup apps, or cloud-based backup services specifically designed for mobile devices.
- **Automatic Backup:** Many mobile devices offer automatic backup options that regularly back up data to the cloud or other storage locations, ensuring that the backup is up-to-date.
- **Restore Options:** In addition to backup, mobile devices often provide easy-to-use restore options to recover data onto a new or reset device.
- **Data Encryption:** To ensure data security, mobile device backup solutions may incorporate encryption techniques to protect data during transmission and storage.

Examples of mobile device backup solutions include iCloud for iOS devices, Google Backup for Android devices, and various third-party backup apps available on app stores.

Back up or restore data on your Android device

- You can back up content, data, and settings from your phone to your Google Account. You can restore your backed up information to the original phone or to some other Android phones.
- You can't use back up when you set up a personal device with a work profile or for work only, or when you set up a company-owned device.
- Restoring data varies by phone and Android version. You can't restore a backup from a higher Android version onto a phone running a lower Android version.

Where your phone data is stored

- Backups are uploaded to Google servers and they're encrypted with your Google Account password. For some data, your phone's screen lock PIN, pattern, or password is also used to encrypt your data so it can be backed up safely. *Your backup data (except what you back up to Google Photos) is erased if:*

- You don't use your device for 57 days You turn off Android backup

Back up content

1. Back up photos and videos.
2. Back up files and folders.

Automatically back up your phone

- To help protect your backed-up data, use a PIN, pattern, or password screen lock, instead of a swipe or Smart Lock.
- You can set up your device to automatically back up your files.

1. Open your device's Settings app.
2. Select **Google Backup**.

Tip: If this is your first time, turn on **Backup by Google One** and follow the onscreen instructions.

3. Tap **Back up now**.

Your Google One backup can take up to 24 hours. When your data is saved, “On” will be below the data types you selected.

Erase after backing up

- After you back up, you can reset your device by erasing everything on it.

Get your data onto a new phone

When you add your Google Account to a phone that's been set up, what you'd previously backed up for that Google Account gets put onto the phone.

- To restore a backed-up account to a reset phone, follow the on-screen steps. For more help, get help from your device manufacturer.
- Your photos and videos are already available in Google Photos. But you can restore the rest of the data you backed up while you set up your new phone for the first time or after a factory reset.
- At setup, to restore your data, follow the on-screen steps. The process can take up to 24 hours.

How Backup handles your data

- The data that backup collects is encrypted in transit.
- Backup sends your data to Google's backup servers and helps you transfer data between devices. Backup collects certain information to perform services on your device. Some of this functionality uses Google Play services.
- For example, backup collects:
Messages, contacts, app settings, and preferences are collected as part of your personal backup.

Assessment questions to the lecture

Qn No	Question	Answer	Bloom's Knowledge Level
1.	What is mobile device backup? a) Storing data on physical backup tapes b) Backing up data to a computer's hard drive c) Copying data to an external storage device d) Saving data from mobile devices to a remote server or	d.	K1

2.	Which of the following data types are commonly backed up from mobile devices? a) System files only b) Applications only c) Contacts, photos, videos, and app data d) Operating system files only	c.	K1
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Students have to prepare answers for the following questions at the end of the lecture

Qn No	Question	Marks	CO	Bloom's Knowledge Level
1	What are the key aspects of Mobile device backup.	2	2	K1
3	Explain mobile device backup in detail.	16	2	K2

Reference Book

Author(s)	Title of the book	Page numbers
	Web resources	

Unit No 4	BACKUP, ARCHIVE AND REPLICATION	Lecture No	7
Topic	Data archive		
Learning Outcome (LO) At the end of this lecture, students will be able to		Bloom's Knowledge Level	
LO1	Define Data Archive.	K1	
LO2	Differentiate Data Archive and Backup.	K2	
LO3	Explain online and offline data storage.	K2	

Data Archive

Organizations make use of these digital assets to generate new revenues, improve service levels, and leverage historical value. This demands frequent and quick retrieval of the fixed contents at multiple locations. An *archive* is a repository where fixed content is placed. Online data availability in the archive can further increase the business value of the referenced information.

Types of Archives

An electronic data archive is a repository for data that has fewer access requirements. It can be implemented as online, nearline, or offline based on the means of access:

- **Online archive:** The storage device is directly connected to the host to make the data immediately available. This is best suited for active archives.
- **Nearline archive:** The storage device is connected to the host and information is local, but the device must be mounted or loaded to access the information.
- **Offline archive:** The storage device is not directly connected, mounted, or loaded. Manual intervention is required to provide this service before information can be accessed.

An archive is often stored on a *write once read many (WORM)* device, such as a CD-ROM. These devices protect the original file from being overwritten. Some tape devices also provide this functionality by implementing file locking capabilities in the hardware or software. Although these devices are inexpensive, they involve operational, management, and maintenance overhead.

Requirements to retain archives have caused corporate archives to grow at a rate of 50 percent or more per year. At the same time, organizations must reduce costs while maintaining required service-level agreements (SLAs). Therefore, it is essential to find a solution that minimizes the fixed costs of the archive's operations and management. Archives implemented using tape devices and optical disks involve many hidden costs.

The traditional archival process using optical disks and tapes is not optimized to recognize the content, so the same content could be archived several times. Additional costs are involved in offsite storage of media and media management. Tapes and optical media are also susceptible to wear and tear. Frequent changes in these device technologies lead to the overhead of converting the media into new formats to enable access and retrieval.

Government agencies and industry regulators are establishing new laws and regulations to enforce the protection of archives from unauthorized destruction and modification. These regulations and standards affect all businesses and have established new requirements for preserving the integrity of information in the archives. These requirements have exposed the hidden costs and shortcomings of the traditional tape and optical media archive solutions.

Data archiving benefits

The greatest benefit of archiving data is it reduces the cost of primary storage. Primary storage is typically expensive, because a storage array must produce a sufficient level of input/output operations per second to meet operational requirements for user read/write activity.

In contrast, archive storage costs less, because it is typically based on a low-performance, high-capacity storage medium. Data archives can be stored on low-cost hard disk drives (HDDs), tape or optical storage that is generally slower than performance disk or flash drives.

Archive storage also reduces the volume of data that must be backed up. Removing infrequently accessed data from the backup data set improves backup and restore performance. Typically, organizations perform data deduplication on data being moved to a lower storage tier, which reduces the overall storage footprint and lowers secondary storage costs.

Data archiving vs. backup

Data archives are not to be confused with data backups, which are copies of data. Although both are considered secondary storage and use a lower-performance, higher-capacity storage medium than primary storage, they serve different purposes. Archives serve a data retention purpose, whereas backups are used for data protection and disaster recovery.

Data archives can be thought of as a data repository for data that is infrequently accessed but still readily available. Backups, on the other hand, are part of a data recovery mechanism that can be used to restore data in the event that it is corrupted or destroyed. Backup data often consists of important information that must be restored quickly when lost or deleted.

Online vs. offline data storage

Data archives take several different forms. Some systems make use of online data storage, which places archive data onto disk systems where it is readily accessible. Archives can be file-based or object storage-based.

Other archival systems use offline data storage in which archive data is written to tape or other removable media using data archiving software, rather than being kept online. Because tape can be removed, tape-based archives consume far less power than disk systems. This translates to lower archive storage costs.

Cloud storage is another possible archive target. Amazon Glacier, for example, is designed for data archiving. This method is inexpensive but requires an ongoing investment. In addition, costs can grow over time as more data is added to the storage cloud. Cloud providers usually store archived data on tape or HDDs.

Assessment questions to the lecture

Qn No	Question	Answer	Bloom's Knowledge Level
1.	What is the primary purpose of data archiving? a) To permanently delete old data b) To organize data for easy retrieval c) To preserve data that is no longer actively used d) To encrypt data for security purposes	b.	K1
2.	_____are not to be confused with data backups, which are copies of data.	Data archives	K1
3.	Archives can be _____	file-based or object storage-based.	K1

Students have to prepare answers for the following questions at the end of the lecture

Qn No	Question	Marks	CO	Bloom's Knowledge Level
1	Define Data Archive.	2	4	K1
2	Differentiate Data Archive and Backup.	2	4	K2
3	Explain online and offline data storage.	8	4	K2

Reference Book

Author(s)	Title of the book	Page numbers
G Somasundaram Alok Shrivastava	EMC Corporation, Information Storage and Management, Wiley, India	190-191

Unit No 4	BACKUP, ARCHIVE AND REPLICATION	Lecture No	8
Topic	Uses of replication and its characteristics		
Learning Outcome (LO) At the end of this lecture, students will be able to		Bloom's Knowledge Level	
LO1	Define Replication.	K1	
LO2	Discuss the uses of replication and its characteristics.	K2	

Replication:

Replication is the process of creating an exact copy of data. Creating one or more replicas of the production data is one of the ways to provide Business Continuity (BC). These replicas can be used for recovery and restart operations in the event of data loss.

The primary purpose of replication is to enable users to have designated data at the right place, in a state appropriate to the recovery need. The replica should provide recoverability and restartability. Recoverability enables restoration of data from the replicas to the production volumes in the event of data loss or data corruption.

It must provide minimal RPO and RTO for resuming business operations on the production volumes, while restartability must ensure consistency of data on the replica. This enables restarting business operations using the replicas.

Replication can be classified into two major categories: local and remote. Local replication refers to replicating data within the same array or the same data center. This chapter provides details about various local replication technologies, along with key steps to plan and design an appropriate local replication solution.

Fast recovery: In the event of a partial failure of the source, or data corruption, a local replica can be used to recover lost data. In the event of a complete failure of the source, the replica can be restored to a different set of source devices.

In either case, this method provides faster recovery and minimal RTO, compared to traditional restores from tape backups. In many instances business operations can be started using the source device before the data is completely copied from the replica.

- **Decision-support activities such as reporting:** Running the reports using the data on the replicas greatly reduces the I/O burden placed on the production device.
- **Testing platform:** A local replica can be used for testing critical business data or applications. For example, when planning an application upgrade, it can be tested using the local replica. If the test is successful, it can be restored to the source volumes.
- **Data migration:** Local replication can also be used for data migration. Data migration may be performed for various reasons, such as migrating from a small LUN to a larger LUN.

Uses of Local Replicas:

One or more local replicas of the source data can be created for various purposes, including the following:

- **Alternate source for backup:** Under normal backup operations, data is read from the production volumes (LUNs) and written to the backup device. This places additional burden on the production infrastructure, as production LUNs are simultaneously involved in production work. As the local replica contains an exact point-in-time (PIT) copy of the source data, it can be used to perform backup operations. This alleviates the backup I/O workload on the production volumes. Another benefit of using local replicas for backup is that it reduces the *backup window* to zero.
- **Fast recovery:** In the event of a partial failure of the source, or data corruption, a local replica can be used to recover lost data. In the event of a complete failure of the source, the replica can be restored to a different set of source devices. In either case, this method provides faster recovery and minimal RTO, compared to traditional restores from tape backups. In many instances business operations can be started using the source device before the data is completely copied from the replica.
- **Decision-support activities such as reporting:** Running the reports using the data on the replicas greatly reduces the I/O burden placed on the production device.
- **Testing platform:** A local replica can be used for testing critical business data or applications. For example, when planning an application upgrade, it can be tested using the local replica. If the test is successful, it can be restored to the source volumes.
- **Data migration:** Local replication can also be used for data migration. Data migration may be performed for various reasons, such as migrating from a small LUN to a larger LUN.

Assessment questions to the lecture

Qn No	Question	Answer	Bloom's Knowledge Level
1.	Replication is the process of creating an exact copy of ____.	Data	K1
2.	Which of the following is NOT a common use case for data replication? a) Disaster recovery b) Load balancing c) Data migration d) Data compression	b.	K1

3	In the event of a partial failure of the source, or data corruption, a local replica can be used to_____.	recover lost data	K1
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Students have to prepare answers for the following questions at the end of the lecture

Qn No	Question	Marks	CO	Bloom's Knowledge Level
1	Define Replication.	2	4	K1
2	Discuss the uses of replication and its characteristics.	8	4	K2
3	Describe the characteristics, types and benefits of replication. (April 2024)	16	4	K2

Reference Book

Author(s)	Title of the book	Page numbers
G Somasundaram Alok Shrivastava	EMC Corporation, Information Storage and Management, Wiley, India	283-285

Unit No 4	BACKUP, ARCHIVE AND REPLICATION	Lecture No	9
Topic	Compute based, and network-based replication		
Learning Outcome (LO) At the end of this lecture, students will be able to		Bloom's Knowledge Level	
LO1	Define Compute based Replication.	K1	
LO2	Define Network based replication.	K2	
LO3	Discuss the Network based replication in detail with neat diagram.	K2	

Local Replication Technologies:

Host-based and storage-based replications are the two major technologies adopted for local replication. File system replication and LVM-based replication are examples of host-based local replication technology.

Storage array-based replication can be implemented with distinct solutions namely, full-volume mirroring, pointer based full-volume replication, and pointer-based virtual replication.

Host-Based Local Replication

In host-based replication, logical volume managers (LVMs) or the file systems perform the local replication process. LVM-based replication and file system (FS) snapshot are examples of host-based local replication.

LVM-Based Replication

In LVM-based replication, logical volume manager is responsible for creating and controlling the host-level logical volume. An LVM has three components: physical volumes (physical disk), volume groups, and logical volumes. A *volume group* is created by grouping together one or more physical volumes. *Logical volumes* are created within a given volume group. A volume group can have multiple logical volumes.

In LVM-based replication, each *logical partition* in a logical volume is mapped to two physical partitions on two different physical volumes, as shown in Figure 13-4. An application write to a logical partition is written to the two physical partitions by the LVM device driver. This is also known as *LVM mirroring*. Mirrors can be split and the data contained therein can be independently accessed. LVM mirrors can be added or removed dynamically.

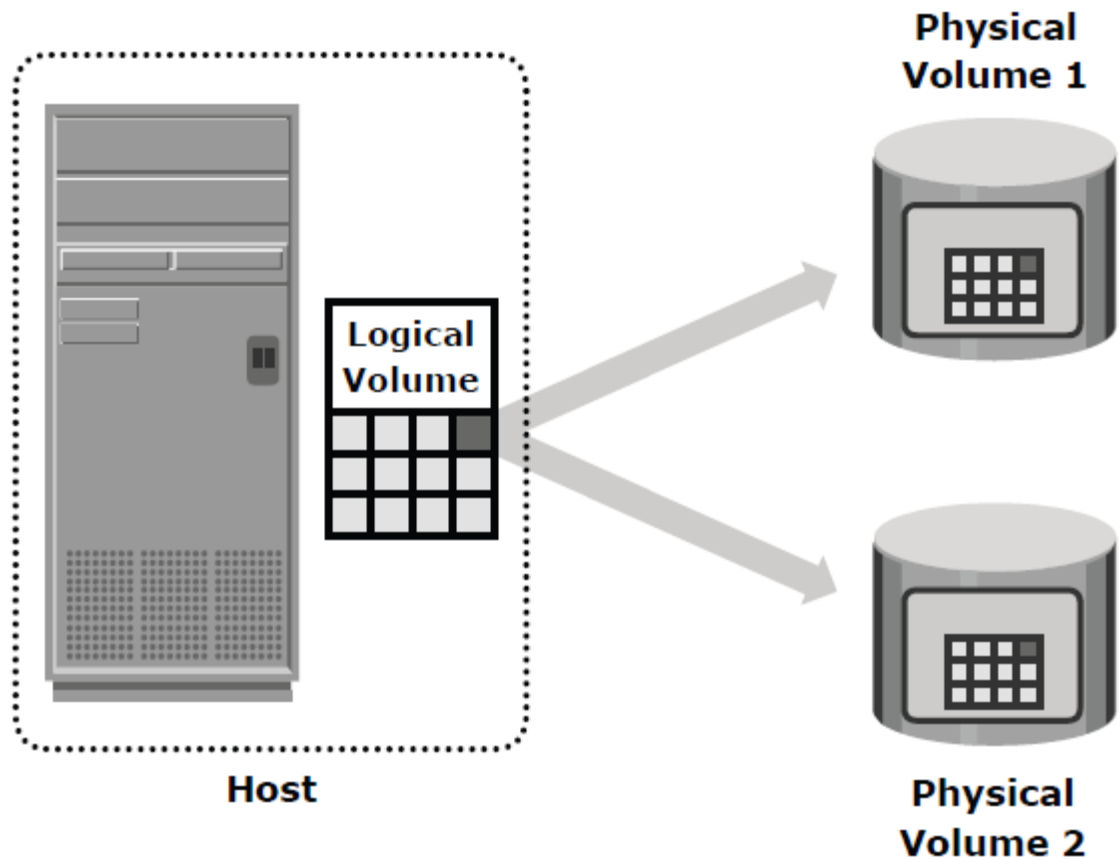


Figure 13-4: LVM-based mirroring

Advantages of LVM-Based Replication

The LVM-based replication technology is not dependent on a vendor-specific storage system. Typically, LVM is part of the operating system and no additional license is required to deploy LVM mirroring.

Limitations of LVM-Based Replication

As every write generated by an application translates into two writes on the disk, an additional burden is placed on the host CPU. This can degrade application performance. Presenting an LVM-based local replica to a second host is usually not possible because the replica will still be part of the volume group, which is usually accessed by one host at any given time.

Tracking changes to the mirrors and performing incremental synchronization operations is also a challenge as all LVMs do not support incremental resynchronization. If the devices are already protected by some level of RAID on the array, then the additional protection provided by mirroring is unnecessary.

This solution does not scale to provide replicas of federated databases and applications. Both the replica and the source are stored within the same volume group. Therefore, the replica itself may become unavailable if there is an error in the volume group. If the server fails, both source and replica are unavailable until the server is brought back online.

File System Snapshot

File system (FS) snapshot is a pointer-based replica that requires a fraction of the space used by the original FS. This snapshot can be implemented by either FS itself or by LVM. It uses Copy on First Write (CoFW) principle. CoFW mechanism is discussed later in the chapter.

When the snapshot is created, a bitmap and a blockmap are created in the metadata of the Snap FS. The bitmap is used to keep track of blocks that are changed on the production FS after creation of the snap. The blockmap is used to indicate the exact address from which data is to be read when the data is accessed from the Snap FS.

Immediately after creation of the snapshot all reads from the snapshot will actually be served by reading the production FS. To read from the Snap FS, the bitmap is consulted. If the bit is 0, then the read is directed to the production FS. If the bit is 1, then the block address is obtained from the blockmap and data is read from that address. Reads from the production FS work as normal.

Assessment questions to the lecture

Qn No	Question	Answer	Bloom's Knowledge Level
1.	_____ is the process of creating replicas of information assets at remote sites a. Remote replication b. Local replication c. Both d. None	a.	K1
2.	Remote replicas help organizations mitigate the risks associated with regionally driven outages resulting from natural or human-made disasters. True False	True	K1

Students have to prepare answers for the following questions at the end of the lecture

Qn No	Question	Marks	CO	Bloom's Knowledge Level
1	Define Host based Replication.	2	4	K1
2	Define Network based Replication.	2	4	K1
3	Discuss the Network based replication in details with neat diagrams	16	4	K2

Reference Book

Author(s)	Title of the book	Page numbers
G Somasundaram Alok Shrivastava	EMC Corporation, Information Storage and Management, Wiley, India	288-290

Unit No 4	BACKUP, ARCHIVE AND REPLICATION	Lecture No	10
Topic	Storage-based replication		
Learning Outcome (LO) At the end of this lecture, students will be able to		Bloom's Knowledge Level	
LO1	Define Storage system based Replication.	K1	
LO2	Discuss the types of Storage system based replication.	K2	

Storage Array–Based Replication

In *storage array-based local replication*, the array operating environment performs the local replication process. The host resources such as CPU and memory are not used in the replication process. Consequently, the host is not burdened by the replication operations. The replica can be accessed by an alternate host for any business operations.

In this replication, the required number of replica devices should be selected on the same array and then data is replicated between source-replica pairs. A database could be laid out over multiple physical volumes and in that case all the devices must be replicated for a consistent PIT copy of the database. Figure 13-5 shows storage array based local replication, where source and target are in the same array and accessed by different hosts.

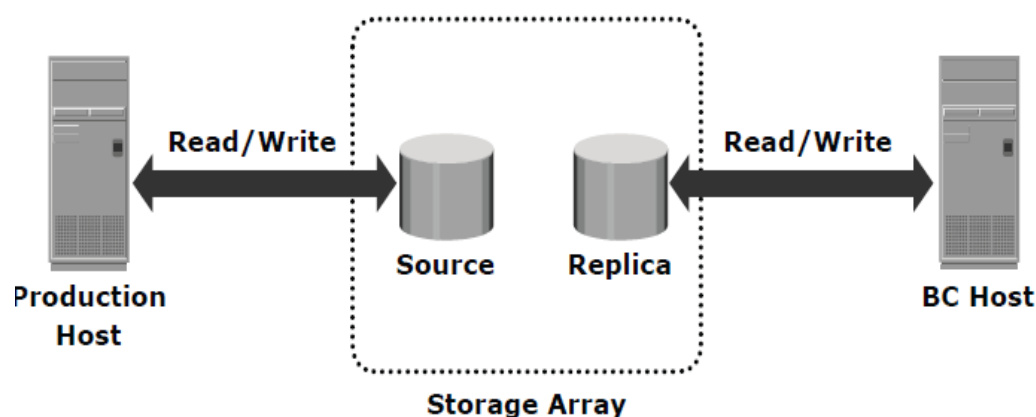


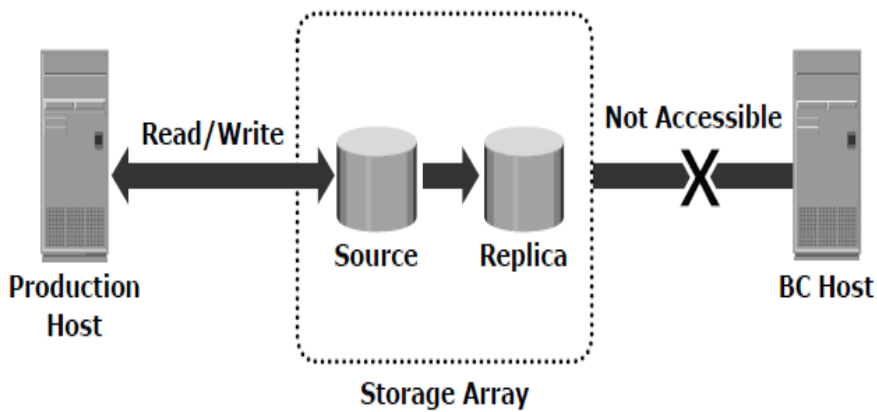
Figure 13-5: Storage array-based replication

Storage array-based local replication can be further categorized as full-volume mirroring, pointer-based full-volume replication, and pointer-based virtual replication. Replica devices are also referred as target devices, accessible by business continuity host.

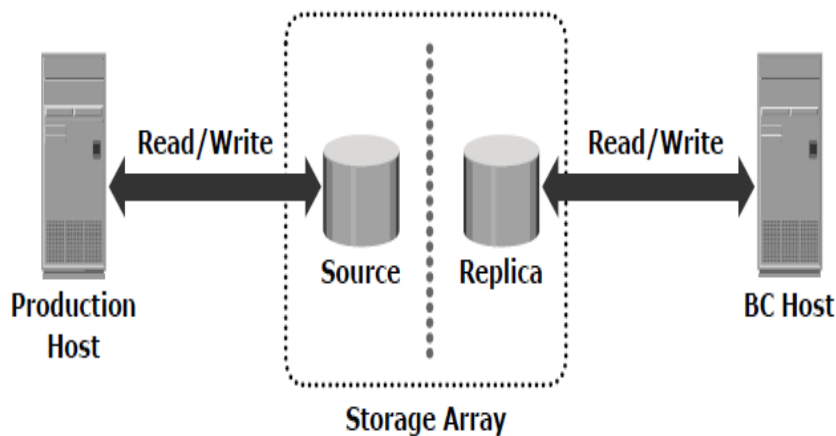
Full-Volume Mirroring

In *full-volume mirroring*, the target is attached to the source and established as a mirror of the source (Figure 13-6 [a]). Existing data on the source is copied to the target. New updates to the source are also updated on the target. After all the data is copied and both the source and the target contain identical data, the target can be considered a mirror of the source. While the target is attached to the source and the synchronization is taking place, the target remains unavailable to any other host. However, the production host can access the source.

After synchronization is complete, the target can be detached from the source and is made available for BC operations. Figure 13-6 (b) shows full-volume mirroring when the target is detached from the source. Notice that both the source and the target can be accessed for read and write operations by the production hosts.



(a) Full volume mirroring with source attached to replica



(b) Full volume mirroring with source detached from replica

Figure 13-6: Full-volume mirroring

After the split from the source, the target becomes a PIT copy of the source. The point-in-time of a replica is determined by the time when the source is detached from the target. For example, if the time of detachment is 4:00 pm, the PIT for the target is 4:00 pm.

After detachment, changes made to both source and replica can be tracked at some predefined granularity. This enables incremental resynchronization (source to target) or incremental restore (target to source). The granularity of the data change can range from 512 byte blocks to 64 KB blocks. Changes are typically tracked using bitmaps, with one bit assigned for each block.

If any updates occur to a particular block, the whole block is marked as changed, regardless of the size of the actual update. However, for resynchronization (or restore), only the changed blocks have to be copied, eliminating the need for a full synchronization (or restore) operation.

This method reduces the time required for these operations considerably. In full-volume mirroring, the target is inaccessible for the duration of the synchronization process, until detachment from the source. For large databases, this can take a long time.

Pointer-Based, Full-Volume Replication

An alternative to full-volume mirroring is *pointer-based full-volume replication*. Like full-volume mirroring, this technology can provide full copies of the source data on the targets. Unlike full-volume mirroring, the target is made immediately available at the activation of the replication session. Hence, one need not wait for data synchronization to, and detachment of, the target in order to access it.

The time of activation defines the PIT copy of source. Pointer-based, full-volume replication can be activated in either Copy on First Access (CoFA) mode or Full Copy mode. In either case, at the time of activation, a protection bitmap is created for all data on the source devices.

Pointers are initialized to map the (currently) empty data blocks on the target to the corresponding original data blocks on the source. The granularity can range from 512 byte blocks to 64 KB blocks or higher. Data is then copied from the source to the target, based on the mode of activation.

In CoFA, after the replication session is initiated, data is copied from the source to the target when the following occurs:

A write operation is issued to a specific address on the source for the first time (see Figure 13-7).

A read or write operation is issued to a specific address on the target for the first time (see Figure 13-8 and Figure 13-9).

When a write is issued to the source for the first time after session activation, original data at that address is copied to the target. After this operation, the new data is updated on the source. This ensures that original data at the point-in-time of activation is preserved on the target. This is illustrated in Figure 13-7.

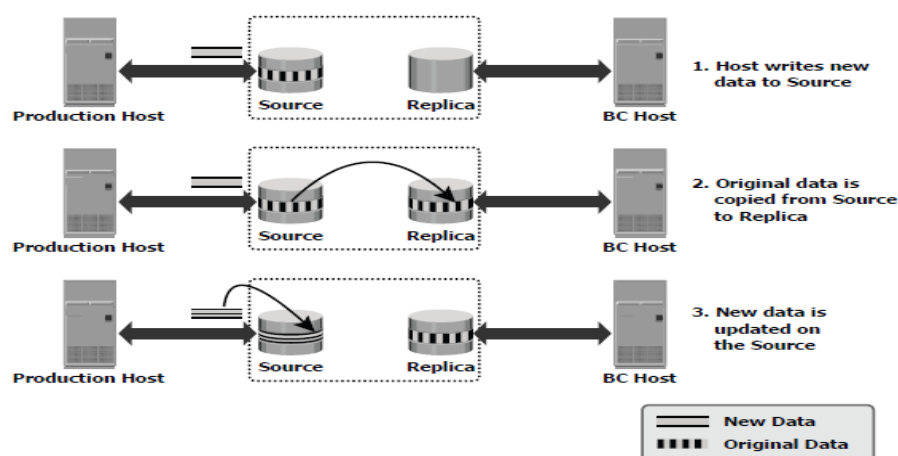


Figure 13-7: Copy on first access (CoFA) – write to source

When a write is issued to the target for the first time after session activation, the original data is copied from the source to the target. After this, the new data is updated on the target. This is illustrated in Figure 13-9. In all cases, the protection bit for that block is reset to indicate that the original data has been copied over to the target.

The pointer to the source data can now be discarded. Subsequent writes to the same data block on the source, and reads or writes to the same data blocks on the target, do not trigger a copy operation (and hence are termed Copy on First Access). If the replication session is terminated, then the target device only has the data that was accessed until the termination, not the entire contents of the source at the point-in-time.

In this case, the data on the target cannot be used for a restore, as it is not a full replica of the source. In Full Copy mode, all data from the source is copied to the target in the background. Data is copied regardless of access. If access to a block that has not yet been copied is required, this block is preferentially copied to the target.

In a complete cycle of the Full Copy mode, all data from the source is copied to the target. If the replication session is terminated now, the target will contain all the original data from the source at the point-in-time of activation. This makes the target a viable copy for recovery, restore, or other business continuity operations.

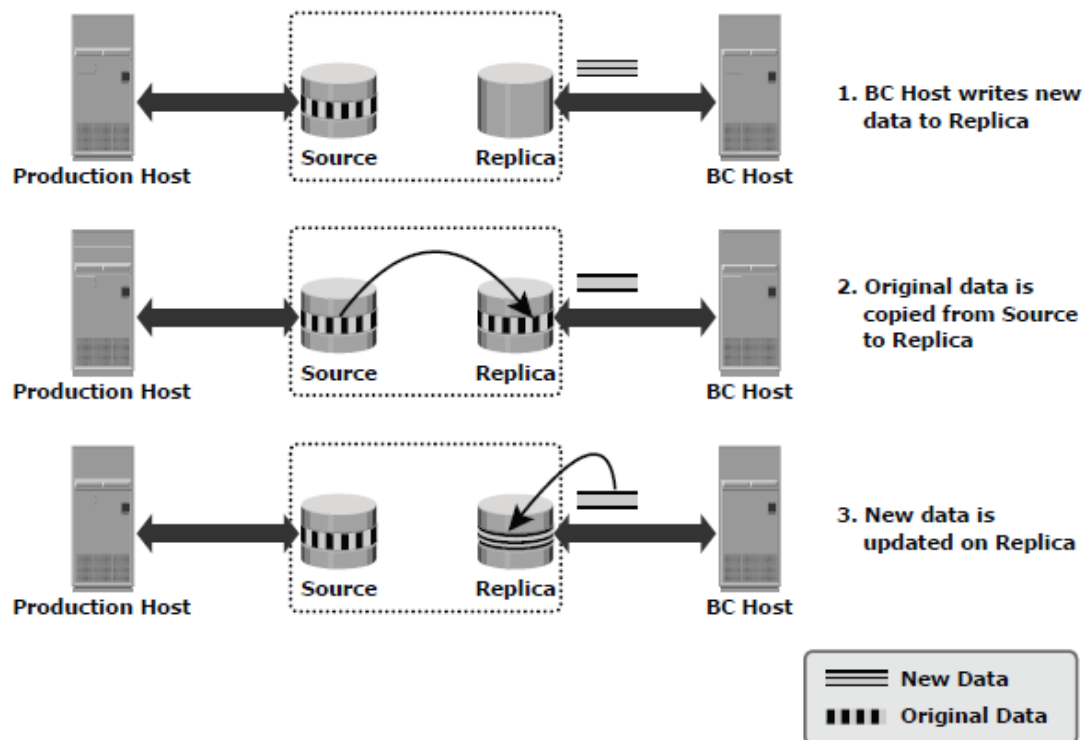


Figure 13-9: Copy on first access (CoFA) – write to target

The key difference between pointer-based, Full Copy mode and full-volume mirroring is that the target is immediately accessible on session activation in Full Copy mode. In contrast, one has to wait for synchronization and detachment to access the target in full-volume mirroring.

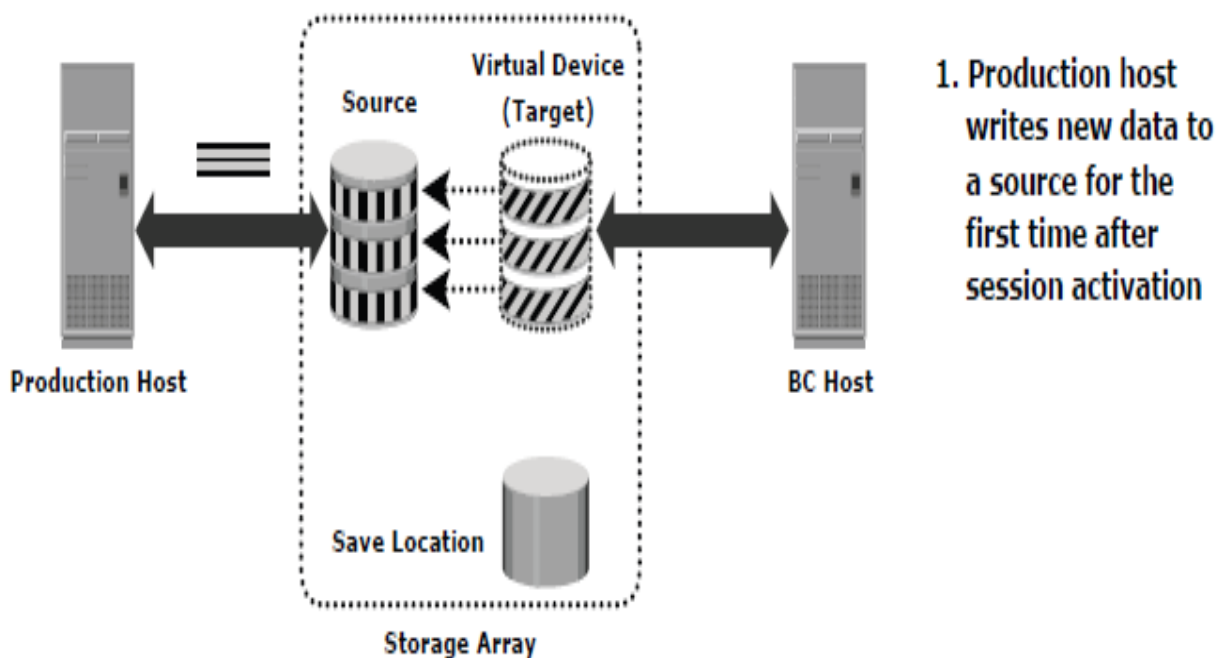
Both the full-volume mirroring and pointer-based full-volume replication technologies require the target devices to be at least as large as the source devices. In addition, full-volume mirroring and pointer-based full-volume replication in Full Copy mode can provide incremental resynchronization or restore capability.

Pointer-Based Virtual Replication

In *pointer-based virtual replication*, at the time of session activation, the target contains pointers to the location of data on the source. The target does not contain data, at any time. Hence, the target is known as a *virtual replica*.

Similar to pointer-based full-volume replication, a protection bitmap is created for all data on the source device, and the target is immediately accessible. Granularity can range from 512 byte blocks to 64 KB blocks or greater. When a write is issued to the source for the first time after session activation, original data at that address is copied to a predefined area in the array. This area is generally termed the *save location*.

The pointer in the target is updated to point to this data address in the save location. After this, the new write is updated on the source. This process is illustrated in Figure 13-10.



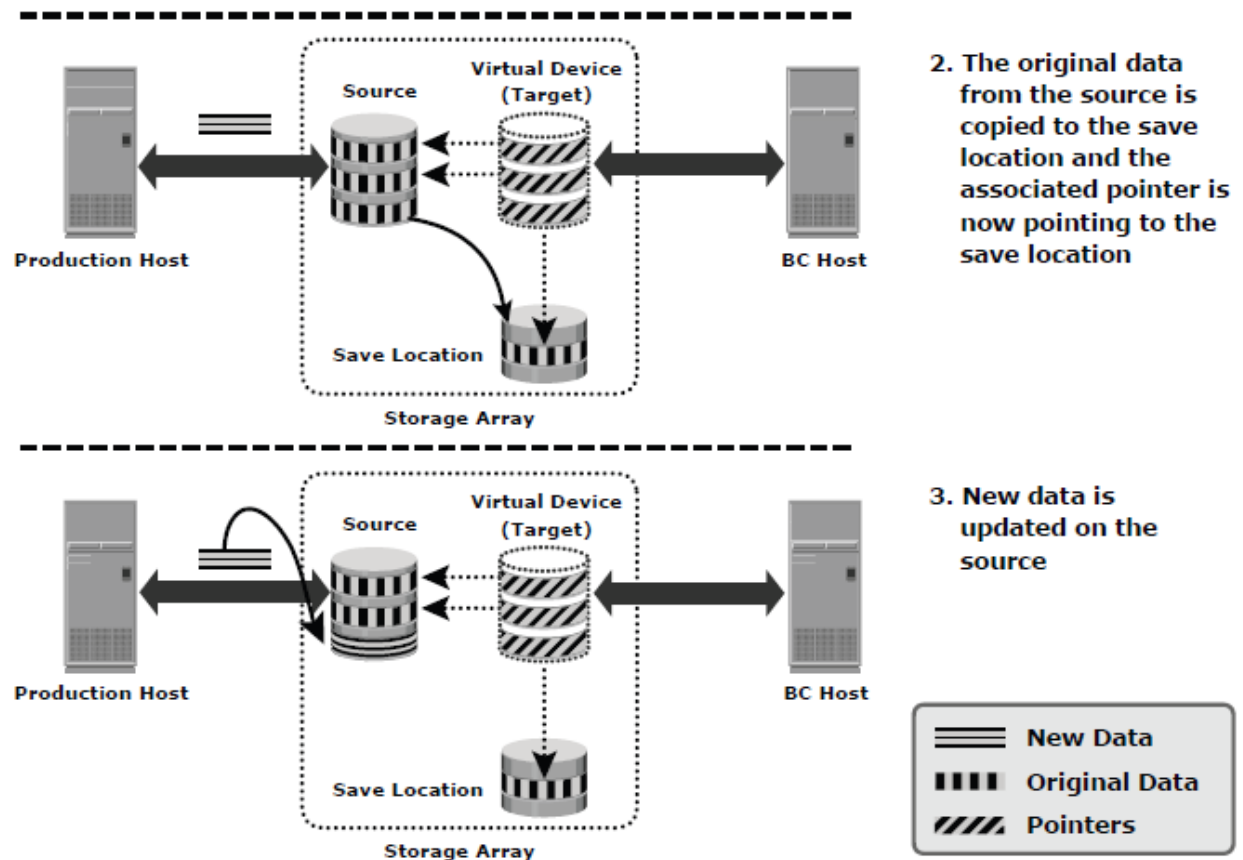


Figure 13-10: Pointer-based virtual replication – write to source

When a write is issued to the target for the first time after session activation, original data is copied from the source to the save location and similarly the pointer is updated to data in save location. Another copy of the original data is created in the save location before the new write is updated on the save location. This process is illustrated in Figure 13-11.

When reads are issued to the target, unchanged data blocks since session activation are read from the source. Original data blocks that have changed are read from the save location. Pointer-based virtual replication uses CoFW technology. Subsequent writes to the same data block on the source or the target do not trigger a copy operation.

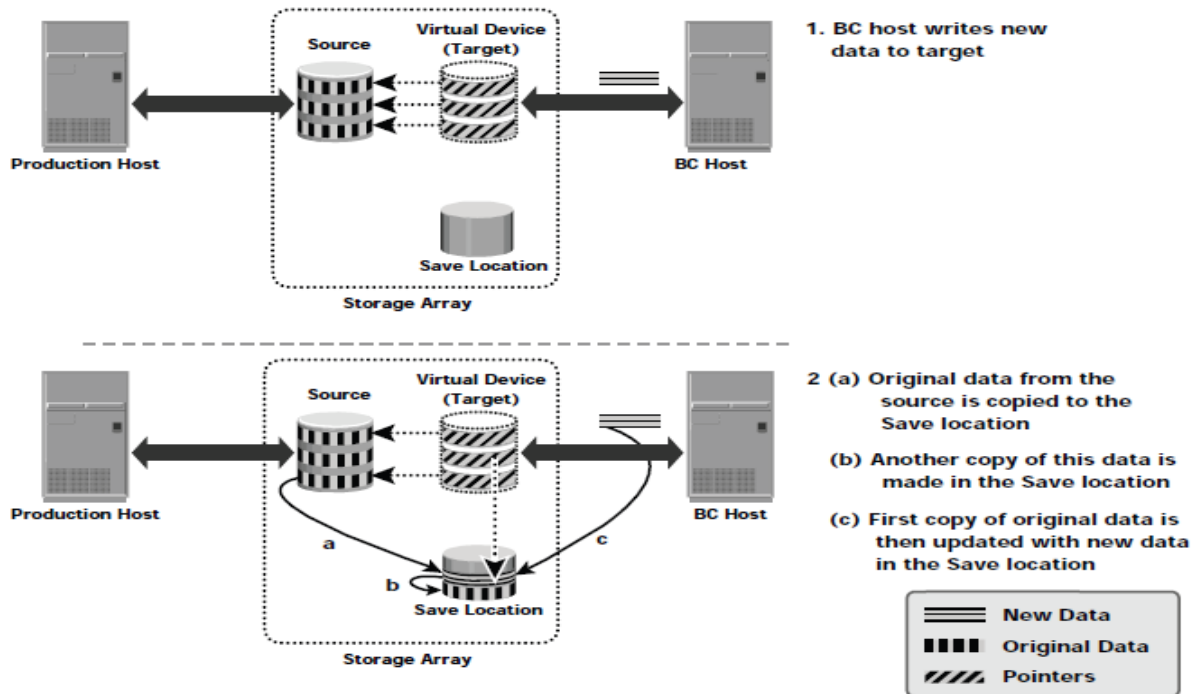


Figure 13-11: Pointer-based virtual replication – write to target

Data on the target is a combined view of unchanged data on the source and data on the save location. Unavailability of the source device invalidates the data on the target. As the target only contains pointers to data, the physical capacity required for the target is a fraction of the source device. The capacity required for the save location depends on the amount of expected data change.

Assessment questions to the lecture

Qn No	Question	Answer	Bloom's Knowledge Level
1.	In, _____the array operating environment performs the local replication process.	<i>storage array-based local replication</i>	K1
2.	In_____, at the time of session activation, the target contains pointers to the location of data on the source.	<i>pointer-based virtual replication</i>	K1

Students have to prepare answers for the following questions at the end of the lecture

Qn No	Question	Marks	CO	Bloom's Knowledge Level
1	Define Storage system based Replication.	2	4	K1
2	Discuss the types of Storage system based replication.	16	4	K2

Reference Book

Author(s)	Title of the book	Page numbers
G Somasundaram Alok Shrivastava	EMC Corporation, Information Storage and Management, Wiley, India	290-297

Unit No 4	BACKUP, ARCHIVE AND REPLICATION	Lecture No	11
Topic	Data migration-Types of Migration-Data migration process.		
Learning Outcome (LO) At the end of this lecture, students will be able to		Bloom's Knowledge Level	
LO1	Describe Data migration.	K1	
LO2	Discuss the types of Data migration.	K2	
LO3	Explain the Data migration process.	K2	

Data Migration

The expression 'data migration' describes the process of moving data within a storage hierarchy as well as moving data from an old storage technology to a new one. In contrast to the storage hierarchy, data migration to new storage technologies is also relevant for small and medium quantities of data

Data migration describes the transfer of data from a source location to a target location. The meaning of 'location' depends on the context and can be anything from storage media, such as magnetic tapes and hard disks, and file systems to applications such as a Document Management System (DMS). During the data migration the copy of the data at the source location is usually invalidated or deleted, whereas the copy of the data at the target is considered active.

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Types of data migrations and their challenges

Data migration is typically performed using one of the following methods:

- **Storage migration** transfers data from one storage device to another. This involves moving blocks of storage and files from storage systems, whether they're on disk, tape or in the cloud. During migration is also an optimal time for organizations to perform data validation and reduction by identifying obsolete or corrupt data.
- **Database migration** moves database files to a new device. This is done when organization changes database vendors, upgrades the database software or moves a database to the cloud. Databases must be backed up before migrating.
- **Application migration** moves an application or program from one environment to another. Application migration typically occurs when an organization switches to another vendor, application or platform. This process is complex because applications interact with other applications, and each one has its own data model. Successful application migration may require using middleware products to bridge technology gaps.
- **Cloud migration** moves data or applications from an on-premises location to the cloud or from one cloud service to another. Cloud migration is a common form of data migration.

Cloud environments provide on-demand flexibility and scalability and reduce the capital expenditure (Capex) for on-premises infrastructures. Public cloud providers offer a variety of services for storage, database and application migrations.

- **Business process migration** moves business applications – including customer, product and operational data -- and processes to a new environment.

During data migrations, teams must pay careful attention to the following challenges:

- **Source data.** Not preparing the source data being moved might lead to data duplicates, gaps or errors when it's brought into the new system or application.
- **Wrong data formats.** Data must be opened in a format that works with the system. Files might not have access controls on a new system if they aren't properly formatted before migration.
- **Mapping data.** When stored in a new database, data should be mapped in a sensible way to minimize confusion.
- **Sustainable governance.** Having a data governance plan in place can help organizations track and report on data quality, which helps them understand the integrity of their data.
- **Security.** Maintaining who can access, edit or remove data is a must for security.

Data migration strategies

Although implementation differs by migration type, there are still **two main strategies** organizations use: ***big bang and trickle migrations***.

❖ **Big bang migrations** transfer all associated data within a set time window. The advantages of creating a migration strategy around this method include lower cost, a quicker move and less complexity. The downside, however, is that big bang migrations require the system to be offline for the entire migration. There's also a risk of losing data if it isn't properly backed up to another location ahead of time.

❖ **Trickle migrations** complete a data migration within phases. During the migration, both old and new systems run at the same time, so there's no downtime, which means there's less risk of losing data. However, trickle migrations are more complicated and need more planning and time to implement properly.

How to create a data migration plan

A data migration project can be challenging because administrators must maintain data integrity and time the project so there's minimal effect on the business and they can keep an eye on costs.

- Having a data migration plan helps to ensure there's minimal disruption and downtime to business processes.
- Factors to consider during a data migration project include how long the migration will take, the amount of downtime required, and the risk to the business due to technical compatibility issues, data corruption or application performance.

Phases of Data Migration

- **Discovery.** This should include considerations such as data sources, destinations, security, cost and which migration strategy to use.
- **Resource assessment.** Identify who will be taking part in the migration.
- **Data inspection.** Examine the data being migrated for data quality, anomalies or duplications. Data should also be backed up.
- **Design.** Data is organized and mapped out for where it's being moved to.
- **Software tools.** Any software that will help in the transition is purchased or created.
- **Migration.** The migration process is initiated.

- **Cleanup.** Old or legacy systems are shut down and decommissioned. A switch to a new DBMS from a different provider — for example, from MySQL to PostgreSQL or from Oracle to MSSQL (so-called *heterogeneous migration*)

Assessment questions to the lecture

Qn No	Question	Answer	Bloom's Knowledge Level
1.	What is data migration? a) Deleting old data from a system b) Transferring data from one system or storage environment to another c) Backing up data for future use d) Encrypting data for security purposes	b.	K1
2.	What is the first step in the data migration process? a) Testing and Validation b) Data Profiling and Cleansing c) Planning and Assessment d) Execution and Monitoring	c.	K1
3.	Which factor is crucial for the success of a data migration project? a) Skipping data validation b) Ignoring stakeholder feedback c) Establishing clear objectives and requirements d) Using outdated migration tools	c.	K1

Students have to prepare answers for the following questions at the end of the lecture

Qn No	Question	Marks	CO	Bloom's Knowledge Level
1	Describe Data migration.	2	2	K1
2	Discuss the types of Data migration.	2	2	K1
3	Explain the Data migration process.	8	2	K2

Reference Book

Author(s)	Title of the book	Page numbers
	Web resources	

Unit No 4	BACKUP, ARCHIVE AND REPLICATION	Lecture No	12
Topic	Disaster recovery as a service		
Learning Outcome (LO) At the end of this lecture, students will be able to		Bloom's Knowledge Level	
LO1	Define DRaaS.	K1	
LO2	Differentiate DRaaS and BaaS.	K2	
LO3	Explain DRaaS.	K2	

Disaster recovery as a service (DRaaS)

DRaaS is a cloud computing service model that allows an organization to back up its data and IT infrastructure in a third party cloud computing environment and provide all the DR orchestration, all through a SaaS solution, to regain access and functionality to IT infrastructure after a disaster.

The as-a-service model means that the organization itself doesn't have to own all the resources or handle all the management for disaster recovery, instead relying on the service provider.

Disaster recovery planning is critical to business continuity. Many disasters that have the potential to wreak havoc on an IT organization have become more frequent in recent years:

- Natural disasters such as hurricanes, floods, wildfires and earthquakes
- Equipment failures and power outages
- Cyber attacks

DRaaS works by replicating and hosting servers in a third-party vendor's facilities versus in the physical location of the organization that owns the workload. The disaster recovery plan is executed on the third-party vendor's facilities in the event of a disaster that shuts down a customer's site.

Organizations may purchase DRaaS plans through a traditional subscription model or a pay-per-use model that allows them to pay only when disaster strikes. As-a-service solutions vary in scope and cost—organizations should evaluate potential DRaaS providers according to their own unique needs and budget.

DRaaS can save organizations money by eliminating the need for provisioning and maintaining an organization's own off-site disaster recovery environment. However, organizations should evaluate and understand service level agreements.

For instance, what happens to recovery times if both the provider and customer are affected by the same natural disaster, such as a large hurricane or earthquake. Different DRaaS providers have different policies on prioritizing which customers get help first in a large regional disaster or allowing customers to perform their own disaster recovery testing.

DRaaS vs. backup as a service (BaaS)

With disaster recovery as a service, the service provider moves an organization's computer processing to its cloud infrastructure in the event of a disaster. This way, the business can continue to operate, even if the original IT infrastructure is totally destroyed or held hostage.

This differs from backup as a service, where only the data, but not the ability to process the data, is duplicated by a third-party provider. Because BaaS is only protecting the data, and not the infrastructure, it is typically less expensive than but most organizations who use BaaS will want to combine it with another disaster recovery tool to ensure business continuity.

Planning for disaster and getting the help you need is something every business needs to consider. Whatever option you choose, a disaster recovery plan is essential for business continuity, and organizations are increasingly turning to DRaaS.

Assessment questions to the lecture

Qn No	Question	Answer	Bloom's Knowledge Level
1.	What is Disaster Recovery as a Service (DRaaS)? a) A physical backup solution for data centers b) A cloud-based disaster recovery solution c) A manual process for restoring data after a disaster d) A software tool for data encryption	b.	K1
2.	Which of the following is a key benefit of Disaster Recovery as a Service (DRaaS)? a) High upfront costs b) Complex implementation c) Reduced recovery time objectives (RTO) d) Limited scalability	c.	K1
3.	What role does data encryption play in Disaster Recovery as a Service (DRaaS)? a) It increases recovery time objectives (RTO) b) It reduces data redundancy c) It ensures data security during replication and storage d) It decreases reliance on cloud resources	c.	K1

Students have to prepare answers for the following questions at the end of the lecture

Qn No	Question	Marks	CO	Bloom's Knowledge Level
1	Define DRaaS.	2	2	K1
2	Differentiate DRaaS and BaaS.	2	2	K1
3	Explain DRaaS in detail.	8	2	K2

Reference Book

Author(s)	Title of the book	Page numbers
	Web resources	